9.7. Morita equivalence

In this section we will discuss the theory of Morita equivalence of finite dimensional algebras. We note that this theory extends with appropiate changes to the infinite dimensional case.

Theorem 9.6.4 shows that a finite abelian category $\mathcal{C}$ can be identified with the category of finite dimensional modules over a finite dimensional algebra B, once we choose a projective generator P of $\mathcal{C}$. However, the projective generator is not unique. Namely, let $P_1, \dots, P_m$ be the indecomposable projective objects of $\mathcal{C}$ (they make sense by Theorem 9.6.4). Then all the projective generators of $\mathcal{C}$ are the objects of the form $P_{\mathbf{n}} := \bigoplus_{i=1}^m n_i P_i$, where $n_i \geq 1$ (check it!). Defining $B_{\mathbf{n}} = B_{\mathbf{n}}(\mathcal{C}) := \mathrm{End}(P_{\mathbf{n}})^{\mathrm{op}}$, we see that the algebras $B_{\mathbf{n}}$ are all the finite dimensional algebras whose category of finite dimensional modules is equivalent to $\mathcal{C}$.

Definition 9.7.1. Finite dimensional algebras A and B are said to be **Morita equivalent** if the abelian categories $A-\mathrm{fmod}$ and $B-\mathrm{fmod}$ are equivalent.

Thus, we obtain that Morita equivalence classes of finite dimensional algebras are the collections of the form $\{B_{\mathbf{n}}(\mathcal{C}), \mathbf{n} \in \mathbb{N}^m\}$.

Note that the dimension of $B_{\mathbf{n}}$ is $\sum c_{ij} n_i n_j$, where (c_{ij}) is the Cartan matrix of $\mathcal{C}$. In particular, the algebra $B = B_{\mathbf{n}}$ with the smallest possible dimension, $\sum c_{ij}$, corresponds to the case when all $n_i = 1$, i.e., to the projective generator $P = \bigoplus_i P_i$. It is easy to see that this algebra is the only one of the $B_{\mathbf{n}}$ for which $B_{\mathbf{n}}/\mathrm{Rad}(B_{\mathbf{n}}) = \bigoplus_i \mathrm{Mat}_{n_i}(k)$ is commutative.

Definition 9.7.2. A finite dimensional algebra B with commutative $B/\mathrm{Rad}(B)$ is called **basic**.

So we have the following corollary of Theorem 9.6.4.

Corollary 9.7.3. *(i) Any finite abelian category $\mathcal{C}$ is equivalent to the category of finite dimensional modules over a unique basic algebra $B = B(\mathcal{C})$.*

(ii) Any finite dimensional algebra A is Morita equivalent to a unique basic algebra $B = B_A$, such that $\dim B_A \leq \dim A$.